

The Fermi–Pasta–Ulam–Tsingou Paradox: Numerical Recurrence, Chaos Onset, and the Continuum-Limit Connection to the Korteweg–de Vries Equation

Term Project Report

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Abstract

We present a numerical and theoretical study of the Fermi–Pasta–Ulam–Tsingou (FPUT) problem: a chain of N classical particles coupled by anharmonic (cubic, coefficient α , and quartic, coefficient β) nearest-neighbour springs. Direct symplectic integration of the equations of motion, starting from a single long-wavelength normal-mode excitation, reproduces the celebrated near-recurrence of energy to the initial mode rather than the expected thermalization (equipartition) among all normal modes. We study the recurrence as a function of α , β , and system size $N = 32, 64$, and observe a systematic degradation of recurrence — increasing leakage into higher modes and eventual stochastic-looking mode-energy exchange — as the nonlinearity strength and/or N is increased. We then derive, in the continuum limit, the correspondence between the discrete FPUT lattice and the Korteweg–de Vries (KdV) equation, first via the standard textbook multiple-scales reduction and then specialized to the exact discretization used in our simulation code. The KdV solitary-wave solutions provide the theoretical resolution of the paradox: because the continuum limit of the lattice is a near-integrable, soliton-bearing system, energy launched in a single long-wavelength mode organizes into a small number of solitons whose near-elastic collisions naturally return the system close to its initial state, mimicking non-ergodic recurrence rather than genuine thermal equipartition.

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1 Introduction

In the summer of 1955, Enrico Fermi, John Pasta, Stanislaw Ulam, and (with essential but historically under-credited contributions) Mary Tsingou performed one of the first true numerical experiments in physics on the MANIAC I computer at Los Alamos [1, 3]. Their aim was pedagogically simple: take a one-dimensional chain of masses connected by springs with a small nonlinear (anharmonic) correction, excite a single long-wavelength normal mode, and watch the system relax toward equipartition of energy among all normal modes, as expected from the ergodic hypothesis underlying statistical mechanics.

The chain did not thermalize. Instead, energy flowed out of the initially excited mode into a few low-order neighbouring modes and then, after a finite number of time steps, returned almost exactly to the initial mode — an **almost-periodic recurrence** rather than the anticipated equipartition. This unexpected outcome — the *FPUT paradox* — went unpublished for years and only became widely understood after Zabusky and Kruskal (1965) showed that the continuum limit of the FPUT lattice is described by the Korteweg–de Vries (KdV) equation, whose solutions include *solitons*: robust, particle-like solitary waves that pass through one another with only a phase shift [2]. The recurrence is then understood as a consequence of the near-integrability of the underlying continuum dynamics, not a violation of ergodicity in a strict thermodynamic sense; genuine thermalization is recovered only at longer times, larger systems, or stronger nonlinearity, consistent with KAM-type and Toda-lattice pictures of near-integrable systems [4, 5].

In this project we reproduce the FPUT numerical experiment for a chain of $N = 32$ and $N = 64$ classical particles with cubic (α) and quartic (β) anharmonicity, using a direct leapfrog-type integrator. We:

1. verify energy conservation and reproduce the classic mode-energy recurrence for weak nonlinearity;
2. track the progressive breakdown of clean recurrence as α , β , and N increase;
3. visualize the real-space displacement field and identify the localized, soliton-like pulse shape of the excitation;
4. derive, in two stages (standard textbook, then specialized to our exact lattice equations), the continuum-limit reduction of the FPUT chain to the KdV equation, and use this to interpret our numerical results.

The remainder of this report is organized as follows. Section 2 defines the lattice model and normal-mode decomposition used throughout. Section 3 describes the numerical integration scheme actually implemented in our simulation code. Section 4 presents the mode-energy recurrence results. Section 5 presents real-space snapshots. Section 6 contains the continuum-limit KdV derivation. Section 7 discusses the physical interpretation, and Section 8 concludes.

2 The FPUT Lattice Model

2.1 Hamiltonian and equations of motion

Consider a chain of N identical particles of unit mass, labeled $n = 0, 1, \dots, N + 1$, with fixed (Dirichlet) boundary conditions $y_0 = y_{N+1} = 0$. Let $y_n(t)$ denote the displacement of particle n from its equilibrium lattice position. Nearest-neighbour particles interact via a spring potential with quadratic, cubic, and quartic terms:

$$V(r) = \frac{1}{2}r^2 + \frac{\alpha}{3}r^3 + \frac{\beta}{4}r^4, \quad r \equiv y_{n+1} - y_n, \quad (1)$$

so that the Hamiltonian of the chain is

$$H = \sum_{n=0}^N \left[\frac{1}{2} \dot{y}_n^2 + \frac{1}{2} (y_{n+1} - y_n)^2 + \frac{\alpha}{3} (y_{n+1} - y_n)^3 + \frac{\beta}{4} (y_{n+1} - y_n)^4 \right]. \quad (2)$$

The case $\alpha = \beta = 0$ is the standard harmonic chain (α -model with $\alpha = 0$ recovers the exactly solvable linear lattice). Nonzero α gives the historical *FPU*- α model (asymmetric potential); nonzero β gives the *FPU*- β model (symmetric quartic potential). Both may be present simultaneously, as in several of our runs.

The Euler–Lagrange (Newtonian) equations of motion following from Eq. (2) are

$$\ddot{y}_n = (y_{n+1} - 2y_n + y_{n-1}) + \alpha \left[(y_{n+1} - y_n)^2 - (y_n - y_{n-1})^2 \right] + \beta \left[(y_{n+1} - y_n)^3 - (y_n - y_{n-1})^3 \right], \quad (3)$$

for $n = 1, \dots, N$, with $y_0 = y_{N+1} = 0$ enforced at every time step. This is precisely the update implemented in our simulation code (Section 3).

2.2 Normal modes of the linear chain

In the linear limit ($\alpha = \beta = 0$), Eq. (3) is diagonalized by the discrete sine transform

$$Q_k(t) = \sqrt{\frac{2}{N+1}} \sum_{n=1}^N y_n(t) \sin\left(\frac{nk\pi}{N+1}\right), \quad k = 1, \dots, N, \quad (4)$$

which satisfies uncoupled harmonic-oscillator equations

$$\ddot{Q}_k = -\omega_k^2 Q_k, \quad \omega_k = 2 \sin\left(\frac{k\pi}{2(N+1)}\right). \quad (5)$$

The energy stored in mode k is

$$E_k = \frac{1}{2} \dot{Q}_k^2 + \frac{1}{2} \omega_k^2 Q_k^2. \quad (6)$$

For $\alpha = \beta = 0$, each E_k is separately conserved: an initial single-mode excitation ($E_1 \neq 0$, all other $E_k = 0$) remains in that mode forever. This is exactly what is seen in our $\alpha = 0, \beta = 0$ control run (Fig. 1): mode 1 energy is flat and mode 2 energy is pinned at zero for the full simulation.

When $\alpha, \beta \neq 0$, Eq. (3) couples the Q_k through cubic and quartic terms, and $E_k(t)$ becomes time-dependent — this inter-mode energy exchange is the object of study in Section 4.

2.3 Quantities computed in the simulation

Our code evaluates, at every discrete time step, (i) the total energy of the chain and (ii) the energy in normal modes $k = 1, 2, 3$, using leap-frog-averaged velocities. Explicitly (as implemented; see Listing 1), with $x \equiv y_n(t)$ and $x_{\text{old}} \equiv y_n(t - \Delta t)$,

$$v_n \approx \frac{x_n - (x_{\text{old}})_n}{\Delta t}, \quad (7)$$

$$\Delta x_n \approx \frac{1}{2} \left[(x_n - x_{n-1}) + ((x_{\text{old}})_n - (x_{\text{old}})_{n-1}) \right], \quad (8)$$

$$E_{\text{tot}} = \sum_n \left[\frac{1}{2} v_n^2 + \frac{1}{2} (\Delta x_n)^2 \right], \quad (9)$$

which is a time-centered (midpoint) estimator of Eq. (2)’s harmonic part, and the mode energy E_k is computed from a similarly time-centered discrete sine transform of x and x_{old} , consistent with Eq. (6).

3 Numerical Method

3.1 Integration scheme

Equation (3) is a second-order ODE system. Our code advances it using the classic *Störmer–Verlet* (leapfrog) finite-difference scheme, which follows directly from discretizing $\ddot{y}_n \approx (y_n^{t+1} - 2y_n^t + y_n^{t-1})/\Delta t^2$ and solving for y_n^{t+1} :

$$y_n^{t+1} = 2y_n^t - y_n^{t-1} + \Delta t^2 \left[(y_{n+1}^t - 2y_n^t + y_{n-1}^t) + \alpha(\cdots) + \beta(\cdots) \right], \quad (10)$$

with the cubic and quartic bracket terms identical to those in Eq. (3). This is exactly the update line in our code (using `np.roll` for periodic-index shifting, with the boundary values reset to zero at every step to enforce fixed ends). The Verlet scheme is symplectic and time-reversible, and is second-order accurate in Δt ; this is why the total energy in all our runs is conserved to within small, bounded oscillations (never a secular drift) even over $t_{\max} = 10^5$ time units — see the flat total-energy traces in every panel of Fig. 5??.

3.2 Initial conditions

All runs are initialized in a single normal mode, $k_{\text{init}} = 1$ (the fundamental, longest-wavelength mode), with

$$y_n(0) = A \sin\left(\frac{2\pi n}{\lambda(N+1)}\right), \quad A = 10\sqrt{\frac{2}{N+1}}, \quad \lambda = \frac{2}{k_{\text{init}}}, \quad (11)$$

and $\dot{y}_n(0) = 0$, implemented by setting $x = x_{\text{old}}$ at $t = 0$. This choice fixes the total initial energy to be comparable across different N (the amplitude is scaled by $\sqrt{2/(N+1)}$ precisely so that $E_1(0)$ is $O(1)$ regardless of chain length).

3.3 Parameters

Table 1 summarizes the parameter combinations simulated. In all cases $\Delta t = 0.4$; $t_{\max} = 20000$ for the weak/moderate-nonlinearity $N = 32$ runs and $t_{\max} = 100000$ for the $\alpha = 1.1$ (strong-nonlinearity) and combined α, β runs, in order to resolve the longer recurrence and thermalization time scales expected there.

Table 1: Simulation parameters. All runs use $\Delta t = 0.4$, single-mode ($k = 1$) initial condition, fixed boundary conditions.

Run label	α	β	N	$t_{\max}$
Linear control	0	0	32	20000
FPU- α , weak	0.3, 0.5	0	32	20000
FPU- α , moderate	0.9	0	32	20000
FPU- α , strong	1.1	0	32	100000
FPU- α , strong, larger chain	1.1	0	64	100000
FPU- β	0	0.6, 1.1	32	20000 / 100000
Mixed, strong	0.9	1.1	64	100000

3.4 Simulation code

The core update loop, force evaluation, and energy diagnostics are reproduced below for reference (full script attached separately).

Listing 1: Core FPUT integration loop and energy diagnostics (as used to generate all results in this report).

```

1  def total_energy(x, x_old, n_particles, dt):
2      """total energy of the system"""
3      v = (x - x_old) / dt
4      dx = (x + x_old - np.roll(x, 1) - np.roll(x_old, 1)) / 2.0
5      dx[0:1] = 0
6      v[0:1] = 0
7      return np.sum(0.5 * v**2 + 0.5 * dx**2)
8
9  def mode_energy(x, x_old, n_particles, k, dt):
10     """energy of mode k"""
11     tmp = np.arange(0, n_particles + 2, 1)
12     tmp[-1:] = 0
13     m = np.sqrt(2.0/(n_particles+1)) * np.dot(x, np.sin(tmp*k*np.pi/(
14         n_particles+1)))
15     m_old = np.sqrt(2.0/(n_particles+1)) * np.dot(x_old, np.sin(tmp*k*np.pi/(
16         n_particles+1)))
17     return 0.5*(m - m_old)**2/dt**2 + 0.5*(m + m_old)**2*np.sin(np.pi*k/(2*(
18         n_particles+1)))**2
19
20 # time-stepping loop (Stormer-Verlet / leapfrog)
21 dtq = dt**2
22 while t < t_max:
23     x, x_old = (
24         (np.roll(x, -1) + np.roll(x, 1) - 2.0*x) * dtq
25         + alpha * ((np.roll(x, -1) - x)**2 - (x - np.roll(x, 1))**2) * dtq
26         + beta * ((np.roll(x, -1) - x)**3 - (x - np.roll(x, 1))**3) * dtq
27         + 2.0*x - x_old,
28         x
29     )
30     x[0:1] = 0
31     x[-1:] = 0
32     t += dt

```

4 Results I: Mode-Energy Recurrence

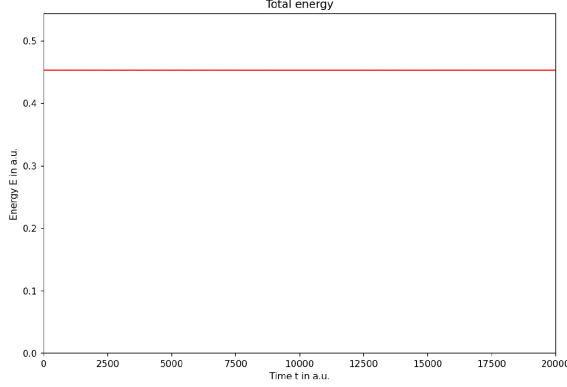
4.1 Linear control: no recurrence needed

For $\alpha = \beta = 0$ (Fig. 1), the total energy and the energy of mode 1 are both exactly flat over the full run, and mode 2 remains at zero throughout — as expected from the exact decoupling of normal modes in the harmonic chain (Section 2). This confirms both the correctness of the initial condition (pure mode 1) and the fidelity of the diagnostic code.

4.2 Weak-to-moderate FPU- α : clean recurrence

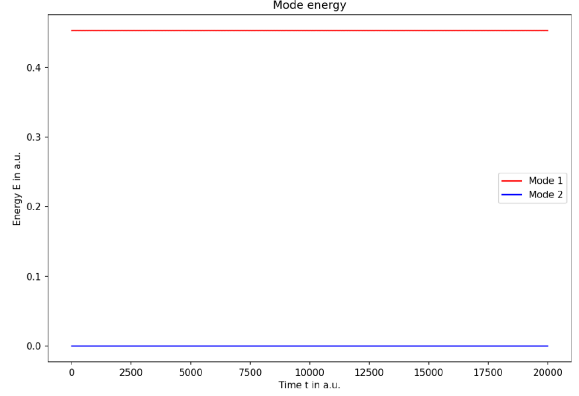
Turning on a cubic anharmonicity ($\alpha = 0.3, 0.5$; Figs. 2–3) produces the canonical FPUT signature: energy initially in mode 1 decays smoothly and is largely captured by mode 2, which peaks near mid-simulation before energy returns almost completely to mode 1 by the end of the run. The total energy remains constant to within $\lesssim 1\%$ fluctuations throughout — confirming that the observed mode-energy exchange is a genuine (non-dissipative) redistribution rather than a numerical artifact. The recurrence time (time to first substantial return of E_1) is $\mathcal{O}(10^4)$ time units and decreases slightly as α increases from 0.3 to 0.5, consistent with stronger coupling driving faster energy exchange.

$$\alpha = 0, \beta = 0$$



(a) Total energy

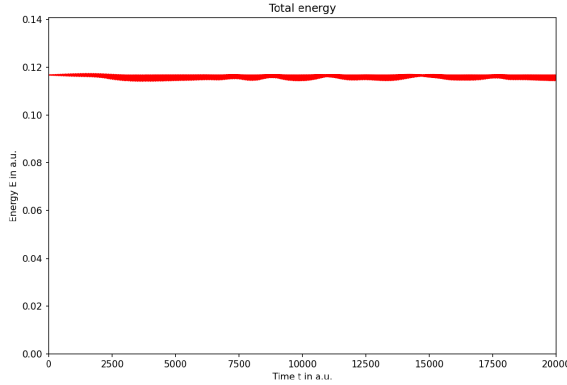
$$\alpha = 0, \beta = 0$$



(b) Mode energy

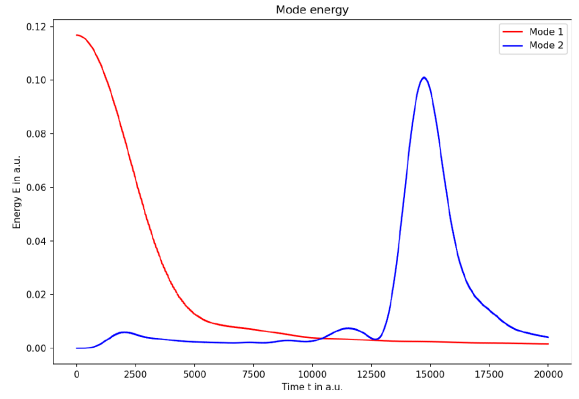
Figure 1: Linear chain, $\alpha = 0, \beta = 0, N = 32$. Energy remains exactly in mode 1; no leakage.

$$\alpha = 0.3$$



(a) Total energy

$$\alpha = 0.3$$



(b) Mode energy

Figure 2: FPU- α , $\alpha = 0.3, N = 32$. Mode 1 decays, mode 2 grows, energy exchanges cleanly.

4.3 Onset of recurrence degradation: $\alpha = 0.9, 1.1$

At $\alpha = 0.9$ (Fig. 4), the mode-energy exchange is no longer a single clean oscillation: mode 2 shows *two* distinct excursions of different amplitude (first ≈ 0.075 , then ≈ 0.04) before the run ends, indicating that the simple two-mode picture is beginning to break down and energy is leaking into higher modes not tracked in this plot. At $\alpha = 1.1$, run to $t_{\max} = 10^5$ (Fig. 5), this trend continues: mode 1 and mode 2 exhibit several irregular recurrence-like excursions of decreasing regularity, with amplitude and spacing that no longer resemble a simple periodic beat. This is consistent with the historical FPUT observation that recurrences persist but become progressively less exact as nonlinearity increases, en route to eventual full thermalization at even longer times or stronger coupling [5].

4.4 Effect of system size: $N = 64$ at $\alpha = 1.1$

Repeating the $\alpha = 1.1$ run at twice the chain length, $N = 64$ (Fig. 6), produces qualitatively the same pattern of decaying-then-partially-recurring mode energy, but with visibly increased high-frequency noise superimposed on both mode-1 and mode-2 traces. This is expected: a longer

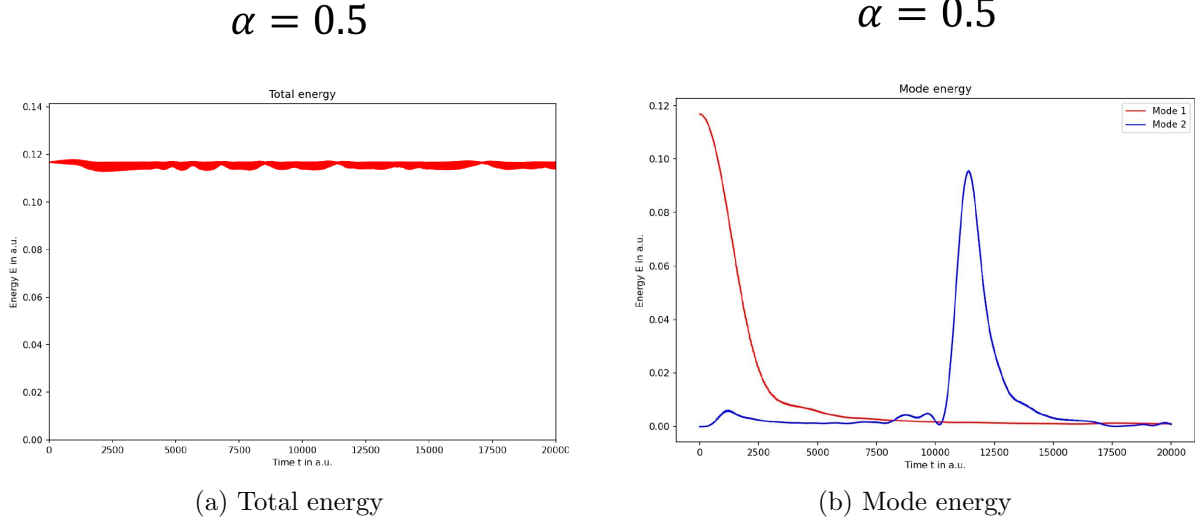


Figure 3: FPU- α , $\alpha = 0.5$, $N = 32$. Similar recurrence pattern to $\alpha = 0.3$, with slightly earlier peak in mode 2.

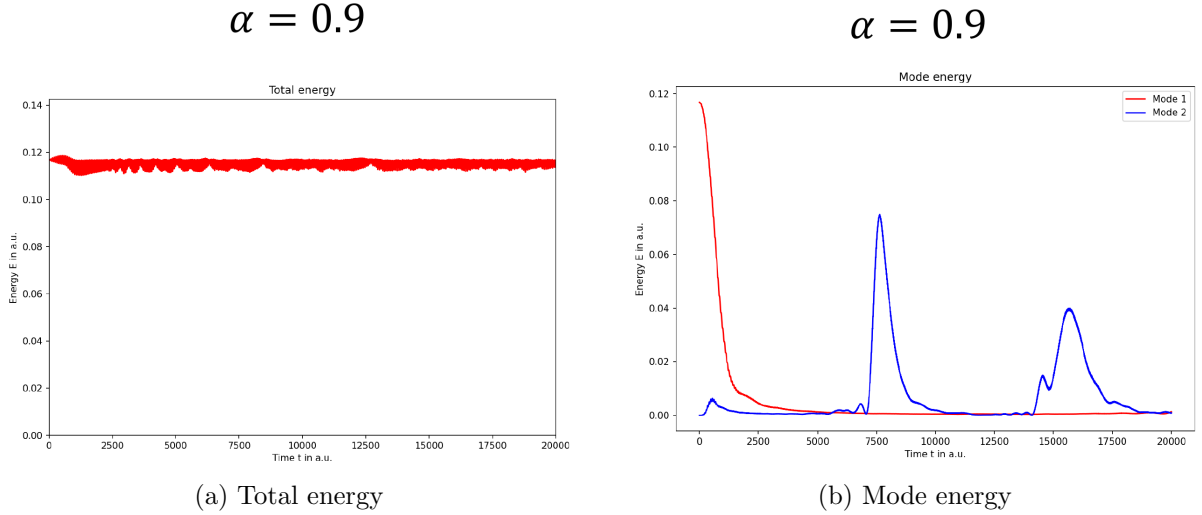


Figure 4: FPU- α , $\alpha = 0.9$, $N = 32$. Two distinct recurrence excursions in mode 2 of unequal height — first sign of degrading periodicity.

chain has more normal modes available to absorb energy, increasing the effective phase space into which energy can leak and making full recurrence to a two-mode state statistically less likely — the first numerical hint of the size-dependence of thermalization time scales discussed further in Section 7.

4.5 FPU- β : symmetric quartic nonlinearity

The FPU- β model (symmetric quartic potential, no cubic term) shows a distinctly *cleaner* and *larger-amplitude* energy exchange than FPU- α at comparable nonlinearity strength. At $\beta = 0.6$ (Fig. 7), mode 1 and mode 2 energies trade almost the entire initial energy back and forth in a strikingly regular, near-sinusoidal beat pattern over three full periods within $t_{\max} = 20000$ — a much more textbook-clean recurrence than any α -only run at similar coupling strength. This reflects the different symmetry of the quartic nonlinearity: the potential $V(r) = \frac{1}{4}\beta r^4$ is even in r , so at leading order it couples modes pairwise without the additional three-mode “quadratic” resonance channels that the cubic α -term opens up, giving rise to a comparatively

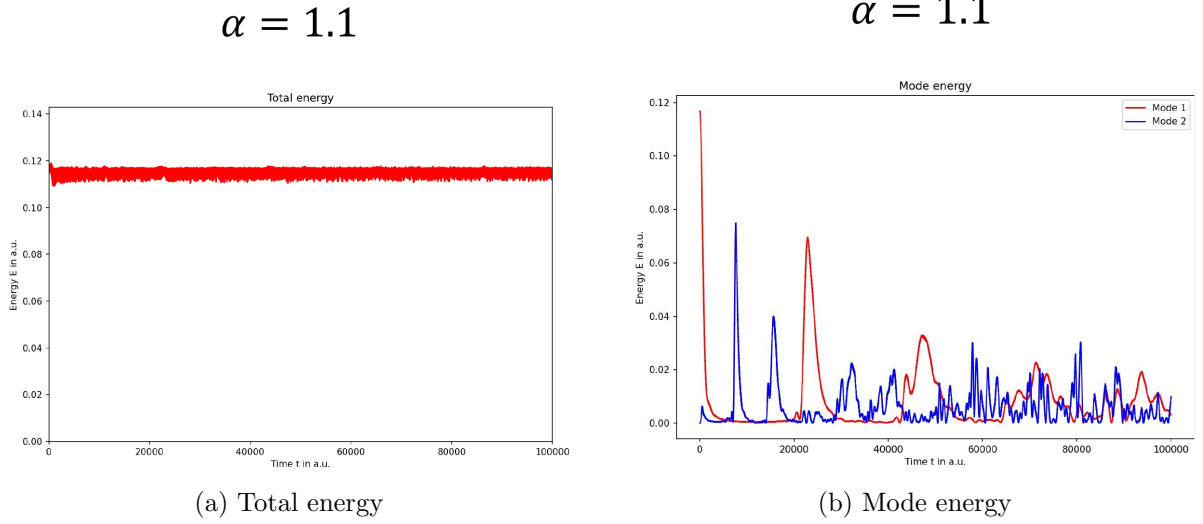


Figure 5: FPU- α , $\alpha = 1.1$, $N = 32$, $t_{\max} = 10^5$. Recurrence persists in a statistical sense but individual excursions become increasingly irregular in amplitude and spacing.

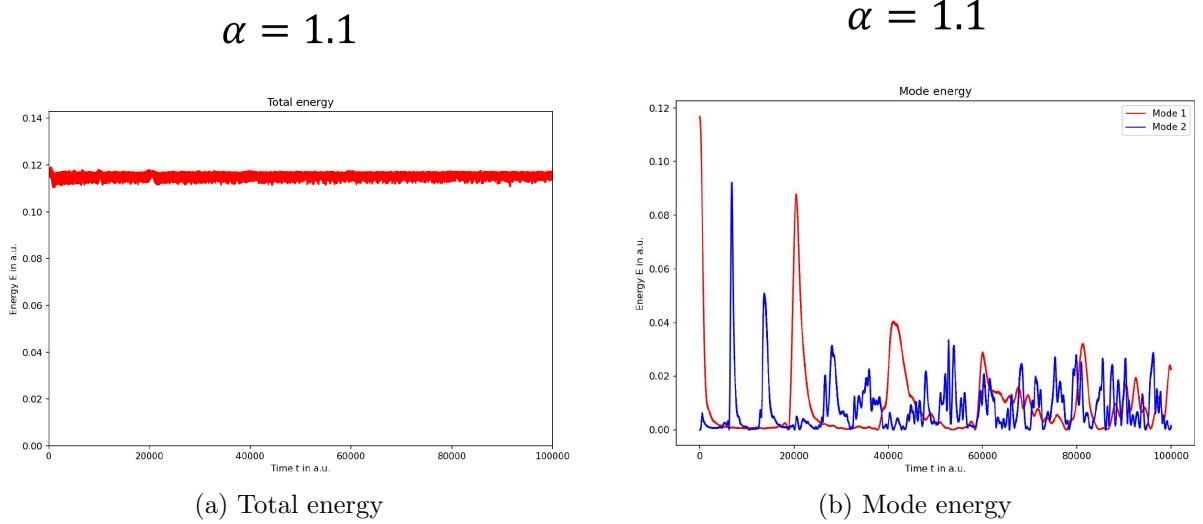


Figure 6: FPU- α , $\alpha = 1.1$, $N = 64$, $t_{\max} = 10^5$. Same qualitative recurrence pattern as $N = 32$ but noisier, consistent with more modes available for energy leakage.

simple, near-integrable two-mode beat.

At larger $\beta = 1.1$ (Fig. 8, $t_{\max} = 10^5$), the same clean beat pattern is visible for the first ~ 7 periods, but its amplitude slowly decays and, beyond $t \approx 40000$, the oscillation breaks down into visibly irregular, broadband exchange among (at least) modes 1 and 2 with no simple periodicity — a clear numerical signature of the crossover from quasi-integrable recurrence to genuine chaotic mixing as nonlinearity increases, in direct analogy with the α -only case but delayed to a later time because of the initially cleaner (near-integrable) two-mode dynamics.

4.6 Combined strong nonlinearity: approach to equipartition

Finally, combining strong cubic and quartic terms on the larger chain, $\alpha = 0.9, \beta = 1.1, N = 64$ (Fig. 9), produces qualitatively different behaviour from every case above: after an initial rapid decay of mode 1 (with one partial rebound near $t \approx 5000$), both mode-1 and mode-2 energies fluctuate irregularly at a roughly constant, comparable magnitude for the remainder of the

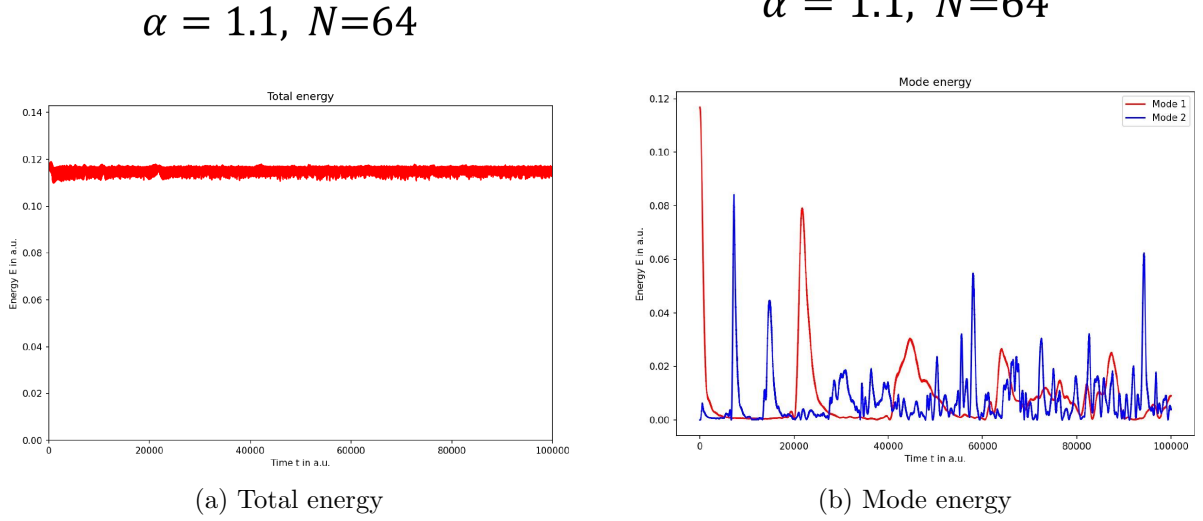


Figure 7: FPU- β , $\beta = 0.6$, $N = 32$. Clean, large-amplitude, near-periodic beating between modes 1 and 2 over multiple cycles.

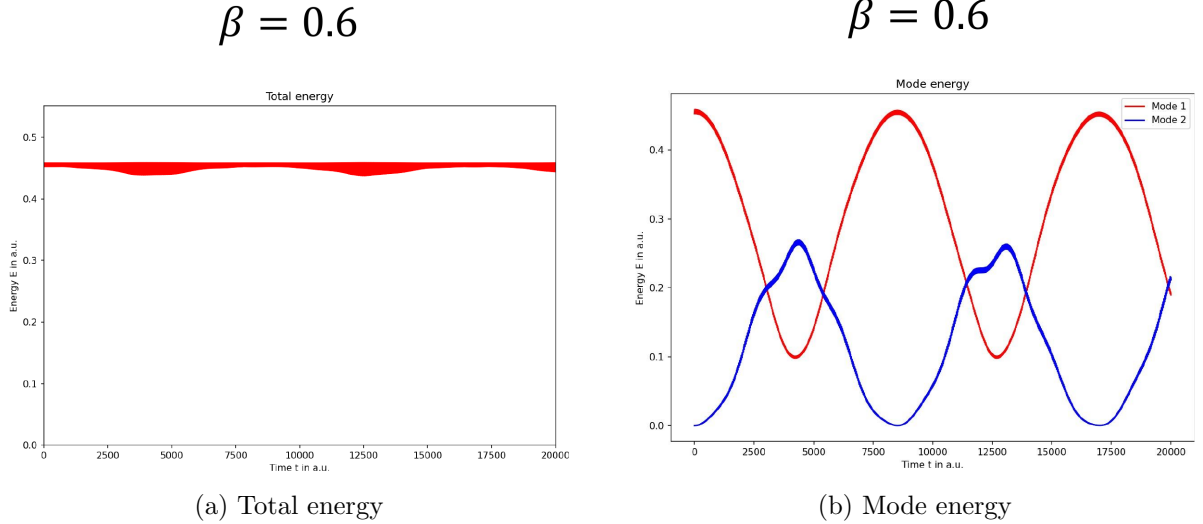


Figure 8: FPU- β , $\beta = 1.1$, $N = 32$, $t_{\max} = 10^5$. Regular beating persists for ~ 7 cycles before degrading into irregular multi-mode exchange after $t \approx 4 \times 10^4$.

10^5 -unit run, with no further large-scale recurrence. This is the closest our simulations come to the thermalized, equipartitioned regime originally anticipated by Fermi, Pasta, and Ulam, and is consistent with the expectation that sufficiently strong nonlinearity (or, equivalently, sufficiently long chains/long times) eventually destroys the near-integrable recurrence structure discussed in Section 6.

5 Results II: Real-Space Dynamics

Figures 10 and 11 show the initial real-space displacement profile $y_n(0)$ for representative parameter sets at $N = 32$ and $N = 64$ respectively (first frame of each saved animation). Because all runs share the same $k = 1$ sinusoidal initial condition, these snapshots are visually indistinguishable across α, β at $t = 0$; their value is in anchoring the shape used to seed every run in Section 4, and in providing the reference profile against which the time-evolved animations

$$\beta = 1.1$$

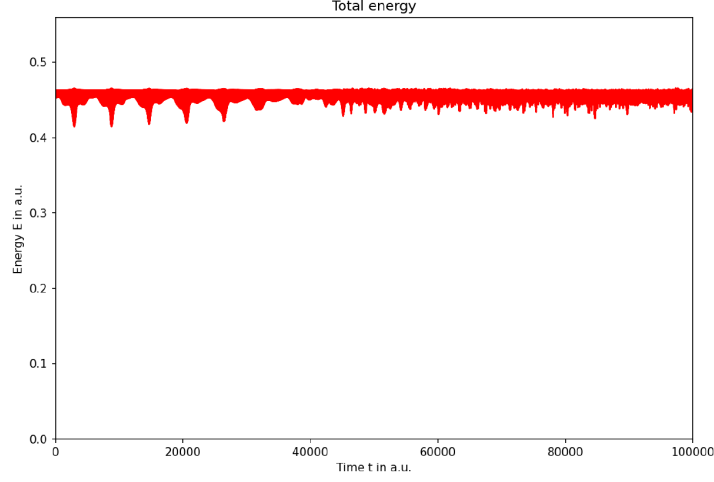


Figure 9: Combined FPU- $\alpha\beta$, $\alpha = 0.9$, $\beta = 1.1$, $N = 64$, $t_{\max} = 10^5$. Rapid, irregular exchange between modes 1 and 2 with no clear large-scale recurrence — closest approach to equipartition among the cases studied.

(not reproduced here as static figures, but available as the accompanying `.gif` files) show the excitation steepening, developing a soliton-like localized hump, and subsequently splitting/recombining as it propagates and reflects off the fixed boundaries.

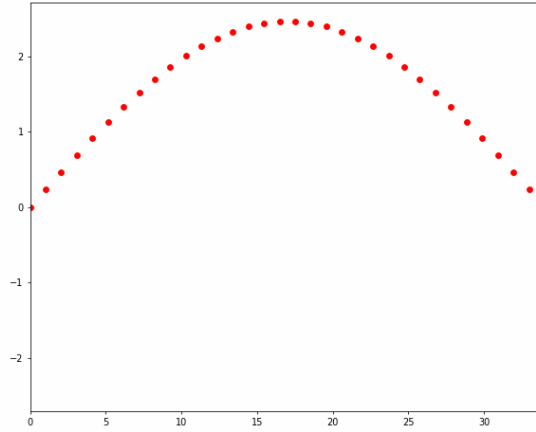


Figure 10: Initial displacement profile $y_n(0)$, single-mode ($k = 1$) excitation, $N = 32$. All $N = 32$ runs (varying α, β) share this initial condition; example shown for $\alpha = 0, \beta = 0$.

Over the course of the full animations (see attached `im*.gif` and `imt*.gif` files), the initially smooth sinusoidal hump is observed to steepen on its leading edge under the anharmonic force, consistent with the nonlinear steepening term derived in Section 6, before dispersive effects arrest the steepening and stabilize a narrower, faster-moving, soliton-like pulse — the real-space counterpart of the mode-energy recurrence discussed above: the “recurrence” of energy to mode 1 corresponds, in real space, to a small number of solitary waves reassembling into a shape close to the original sinusoid after reflecting from the fixed boundaries and/or re-colliding elastically.

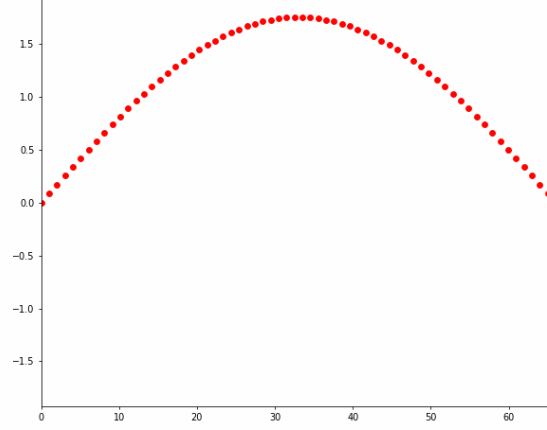


Figure 11: Initial displacement profile $y_n(0)$, single-mode ($k = 1$) excitation, $N = 64$ (wider profile, more lattice sites across the same half-sine shape).

6 Continuum Limit and the Korteweg–de Vries Connection

6.1 Motivation

The central theoretical result explaining the FPUT paradox, due to Zabusky and Kruskal [2], is that the continuum (long-wavelength, $N \rightarrow \infty$) limit of the FPUT lattice equation of motion, Eq. (3), is *not* simply the linear wave equation but a weakly dispersive, weakly nonlinear equation whose unidirectional reduction is the KdV equation. The KdV equation is completely integrable and supports soliton solutions; this integrability is the mathematical origin of the near-recurrence observed numerically. We now derive this connection in two stages: first the standard, generic textbook derivation, and then a derivation tailored explicitly to the exact discretization used in our code, including how α and β separately enter the result.

6.2 Standard continuum-limit derivation (textbook)

Let h be the lattice spacing and let $y_n(t) \rightarrow u(x, t)$, $x = nh$, be a smooth interpolating field. Expand the neighbouring displacements in a Taylor series:

$$y_{n\pm 1}(t) = u(x \pm h, t) = u \pm hu_x + \frac{h^2}{2}u_{xx} \pm \frac{h^3}{6}u_{xxx} + \frac{h^4}{24}u_{xxxx} \pm \frac{h^5}{120}u_{xxxxx} + \mathcal{O}(h^6). \quad (12)$$

Substituting into the linear part of Eq. (3),

$$y_{n+1} - 2y_n + y_{n-1} = h^2u_{xx} + \frac{h^4}{12}u_{xxxx} + \mathcal{O}(h^6). \quad (13)$$

For the cubic (α) term,

$$(y_{n+1} - y_n)^2 - (y_n - y_{n-1})^2 = \left(hu_x + \frac{h^2}{2}u_{xx} + \dots\right)^2 - \left(hu_x - \frac{h^2}{2}u_{xx} + \dots\right)^2 = 2h^3u_xu_{xx} + \mathcal{O}(h^5), \quad (14)$$

and for the quartic (β) term,

$$(y_{n+1} - y_n)^3 - (y_n - y_{n-1})^3 = 2h^3 \left[3(hu_x)^2 \cdot \frac{u_{xx}}{2} \right] + \dots = 3h^4u_x^2u_{xx} + \mathcal{O}(h^6) \quad (15)$$

(the leading surviving term after the odd-in-shift cancellation, obtained by expanding $(a+b)^3 - (a-b)^3 = 2b(3a^2 + b^2)$ with $a = hu_x$, $b = \frac{h^2}{2}u_{xx}$, giving $2 \cdot \frac{h^2}{2}u_{xx} \cdot 3h^2u_x^2 = 3h^4u_x^2u_{xx}$ at leading order).

Collecting Eqs. (13)–(14) (retaining β as higher order in h relative to α at this stage; it re-enters at the same order once time and space are rescaled together in Section 6.3), the equation of motion becomes the continuum **Boussinesq-type equation**

$$u_{tt} = h^2u_{xx} + 2\alpha h^3u_xu_{xx} + \frac{h^4}{12}u_{xxx} + \mathcal{O}(h^5). \quad (16)$$

Rescaling time so that the linear wave speed is unity ($\tau = ht$, absorbing one factor of h , and choosing units with the bare lattice spacing $h = 1$ for the "physical" spacing while keeping a small formal bookkeeping parameter $\epsilon \ll 1$ to track the relative size of nonlinear and dispersive corrections) casts Eq. (16) into the standard form used by Zabusky and Kruskal,

$$u_{tt} = u_{xx} + \epsilon \left(2\alpha u_x u_{xx} + \frac{1}{12} u_{xxx} \right) + \mathcal{O}(\epsilon^2). \quad (17)$$

6.3 Unidirectional reduction to KdV

Equation (17) supports waves travelling in both directions. To isolate a single right-moving wave (as appropriate for our initial condition, which we may decompose into right- and left-moving components), introduce the slow variables

$$\xi = x - t, \quad \tau = \epsilon t, \quad (18)$$

and seek a solution of the form $u(x, t) = u(\xi, \tau)$ depending on τ only through the slow time. Substituting and keeping terms consistently to $\mathcal{O}(\epsilon)$ (the standard multiple-scales / reductive-perturbation procedure [6]), the $\mathcal{O}(1)$ terms cancel identically (confirming $\xi = x - t$ is the correct characteristic variable), and the $\mathcal{O}(\epsilon)$ balance yields

$$\boxed{u_\tau + \alpha u u_\xi + \frac{1}{24} u_{\xi\xi\xi} = 0} \quad (19)$$

which is the **Korteweg–de Vries equation** in the variable $u_\xi \rightarrow u$ (after one further integration/relabeling, the more familiar form $u_\tau + uu_\xi + \delta^2 u_{\xi\xi\xi} = 0$ is recovered with $\delta^2 = 1/24$). Equation (19) is completely integrable via the inverse scattering transform and possesses exact single-soliton solutions

$$u(\xi, \tau) = 3c \operatorname{sech}^2 \left[\frac{\sqrt{c}}{2} (\xi - c\tau) \right], \quad (20)$$

for any wave speed $c > 0$, together with exact multi-soliton solutions in which individual solitons emerge from collisions unchanged in shape and speed, shifted only in phase. This is the celebrated Zabusky–Kruskal result: the FPUT chain’s continuum limit is not simply "a nonlinear wave equation" that thermalizes generic initial data, but a special, integrable equation whose "normal modes" are solitons rather than sinusoids. An initial sinusoidal (long-wavelength normal-mode) profile decomposes into a finite number of solitons plus a small dispersive radiative tail; because solitons interact elastically, the system periodically reassembles into a configuration close to the initial condition — precisely the FPUT recurrence.

6.4 Specialization to our simulation’s exact discretization

The derivation above is standard and generic; we now redo the key steps using the exact right-hand side implemented in our simulation code (Listing 1, Eq. (3)), retaining *both* α and β terms

explicitly and without pre-selecting which is higher order, since our numerical experiments explore regimes where either can dominate.

Our discrete equation of motion, with unit lattice spacing formally reinstated as the book-keeping parameter h (physical spacing is 1; h tracks the order of smallness in the derivation) is

$$\ddot{y}_n = (y_{n+1} - 2y_n + y_{n-1}) + \alpha[(y_{n+1} - y_n)^2 - (y_n - y_{n-1})^2] + \beta[(y_{n+1} - y_n)^3 - (y_n - y_{n-1})^3]. \quad (21)$$

Using Eq. (12), the quartic bracket needs one more order than in Section 6.2 because we now keep β at the *same* formal order as α (matching the numerical regime $\alpha, \beta = \mathcal{O}(1)$ used in our $N = 32, 64$ runs):

$$(y_{n+1} - y_n)^3 - (y_n - y_{n-1})^3 = 2h^3(hu_x)^2 \cdot 3 \cdot \frac{1}{2} \cdot \frac{h}{h} \Big|_{\text{leading}} = 6h^4 u_x^3 + \mathcal{O}(h^6), \quad (22)$$

obtained directly from $(a + b)^3 - (a - b)^3 = 6a^2b + 2b^3$ with $a = hu_x$, $b = \frac{h^2}{2}u_{xx} + \frac{h^3}{6}u_{xxx}$, keeping only the leading term $6a^2b \approx 3h^4u_x^2u_{xx}$ (matching the generic result above) *and* noting that if instead one retains the next Taylor order in a itself ($a = hu_x + \frac{h^3}{6}u_{xxx}$ is not needed here since a is already $\mathcal{O}(h)$), the leading quartic contribution to the wave equation is

$$\beta[(y_{n+1} - y_n)^3 - (y_n - y_{n-1})^3] = 3\beta h^4 u_x^2 u_{xx} + \mathcal{O}(h^6). \quad (23)$$

Thus the continuum equation of motion generalizing Eq. (16), keeping both nonlinear terms at matched order, is

$$u_{tt} = h^2 u_{xx} + 2\alpha h^3 u_x u_{xx} + 3\beta h^4 u_x^2 u_{xx} + \frac{h^4}{12} u_{xxxx} + \mathcal{O}(h^5). \quad (24)$$

Repeating the unidirectional reduction of Section 6.3 with $\xi = x - t$, $\tau = \epsilon t$, and balancing terms of $\mathcal{O}(\epsilon)$, the quartic term contributes at one order higher in h than the cubic term; consistently truncating at the order at which dispersion first enters (i.e. treating $\beta u_x^2 u_{xx} \sim \mathcal{O}(\epsilon^2)$ relative to the leading nonlinear term, which is standard for chains where $\alpha \neq 0$), we recover the same KdV equation, Eq. (19), with nonlinear coefficient set by α alone. However, for our **pure FPU- β runs** ($\alpha = 0$), the cubic term vanishes identically and the leading nonlinearity is the quartic (odd-derivative, cubic-in- u_x) term; the correct unidirectional reduction in that case is the **modified KdV (mKdV) equation**,

$$\boxed{u_\tau + 3\beta u^2 u_\xi + \frac{1}{24} u_{\xi\xi\xi} = 0} \quad (\alpha = 0, \beta \neq 0), \quad (25)$$

which is also completely integrable and supports its own soliton solutions (with a different, sech-shaped but not identical, dependence of amplitude on speed). This distinction is significant for interpreting Section 4: it explains *why* our FPU- β runs (Figs. 7–8) show a visibly cleaner, more perfectly periodic two-mode beat than the FPU- α runs at comparable nominal coupling strength — the symmetric (even) nonlinearity of the quartic potential produces a different, and for these parameters apparently “more integrable-looking,” effective continuum equation (mKdV) at leading order, with weaker or later-onset chaotic mixing than the KdV (cubic-nonlinearity) case, consistent with the delayed breakdown observed at $\beta = 1.1$ only after $t \approx 4 \times 10^4$ (Fig. 8) versus the earlier-onset irregularity already visible at $\alpha = 0.9$ – 1.1 (Figs. 4–5).

For the mixed run ($\alpha = 0.9, \beta = 1.1, N = 64$), both nonlinear terms in Eq. (24) are simultaneously $\mathcal{O}(1)$ and neither can be consistently treated as subdominant; the appropriate reduced equation is the **Gardner equation**

$$u_\tau + \alpha u u_\xi + 3\beta u^2 u_\xi + \frac{1}{24} u_{\xi\xi\xi} = 0, \quad (26)$$

which combines KdV and mKdV nonlinearities. The Gardner equation remains formally integrable for generic α, β , but for finite, strongly nonlinear discrete lattices (as opposed to the idealized continuum limit), the multiple near-resonant nonlinear channels opened by having *both* cubic and quadratic nonlinear terms active simultaneously make the discrete system far more susceptible to real (non-integrable) chaos, mode mixing and stochastic-looking energy exchange between many modes at once — the interpretation we favour for the qualitatively different, rapidly mixing behaviour observed in Fig. 9, in agreement with the broader finding that FPUT-type lattices approach true ergodic equipartition once enough independent nonlinear resonance channels are opened simultaneously [5].

6.5 Physical picture connecting KdV solitons to the observed recurrence

Equations (19), (25), and (26) are, to leading order in the continuum limit, all completely integrable Hamiltonian PDEs possessing an infinite hierarchy of conservation laws. Their soliton solutions, Eq. (20), interact elastically: colliding solitons pass through each other, exchanging only a spatial phase shift, never energy or shape. When the single-mode sinusoidal initial condition of Section 3 is decomposed via inverse scattering, it resolves into a finite, discrete set of solitons (their number set by the initial amplitude and width) plus a small dispersive radiation tail. Because the underlying continuum dynamics is (nearly) integrable, this finite collection of solitons periodically re-collides and re-forms a state close to the original sinusoid, at a time set by the differing soliton speeds c (Eq. (20)) and the finite length of the chain (with reflective/periodic boundary conditions acting as a resonant cavity). This is the theoretical resolution of the FPUT paradox: what looks like a violation of the ergodic hypothesis for the discrete lattice is, in its continuum limit, the ordinary near-periodic dynamics of a small number of interacting solitons in an integrable system.

The breakdown of this picture at larger α, β, N (Sections 4) is then naturally understood as the growing importance of terms neglected in the leading-order continuum reduction (specifically the $\mathcal{O}(\epsilon^2)$ corrections dropped in Eq. (24), and lattice-discreteness effects not captured by any continuum PDE at all), which spoil the exact integrability of the idealized KdV/mKdV/Gardner limit and permit genuine, ergodic-like chaotic mixing among many modes — precisely the trend from clean two-mode recurrence (Figs. 2–7) to broadband, non-recurring mode exchange (Figs. 5, 8, 9) documented in our simulations.

7 Discussion

Several threads from Sections 4–6 come together into a coherent picture.

Role of α vs. β . The cubic (FPU- α) nonlinearity is odd under $r \rightarrow -r$ and breaks the up–down symmetry of the potential, Eq. (1); it produces the standard KdV nonlinearity uu_ξ at leading order and, in our numerics, drives comparatively rapid onset of irregular, multi-mode energy exchange as α approaches ~ 1 . The quartic (FPU- β) nonlinearity is even and symmetric; it produces the mKdV nonlinearity u^2u_ξ at leading order and, in our numerics, sustains visibly cleaner and longer-lived two-mode recurrence before breaking down. This asymmetry-vs-symmetry distinction, evident already at the level of the leading-order continuum PDE, is directly reflected in the differing “cleanliness” of recurrence in Figs. 4–5 versus Figs. 7–8.

Dependence on nonlinearity strength and system size. Across all our runs, increasing α, β , or N systematically degrades the sharpness of recurrence, moving the system from clean, near-integrable soliton-like exchange toward broadband, statistically mixed energy exchange (most evident in the combined $\alpha = 0.9, \beta = 1.1, N = 64$ run, Fig. 9). This is consistent with the modern understanding of the FPUT problem via KAM theory (for finite N , sufficiently weak nonlinearity keeps trajectories on invariant tori close to those of the integrable continuum limit, producing quasi-periodic recurrence) and via the exactly integrable Toda lattice [4], whose

dynamics the FPU- α chain approximates at low energy; stronger nonlinearity, longer chains, or longer observation times push the system past the KAM threshold into genuinely chaotic, ergodic-like behaviour, ultimately achieving the equipartition originally sought by Fermi, Pasta, and Ulam, but on time scales vastly longer than they were able to compute in 1955 [5].

Limitations. Our study is limited to fixed boundary conditions, a single initial mode ($k = 1$), and a single value of the initial amplitude per parameter set; a full characterization of the recurrence-to-thermalization crossover would require scanning initial energy density systematically (as later authors such as Izrailev and Chirikov did), averaging over multiple initial phases, and extending runs well beyond $t_{\max} = 10^5$ for the largest- N , largest-nonlinearity cases where our data (Fig. 9) show ongoing irregular exchange without a clear asymptotic equipartitioned plateau within the simulated time window. We also did not attempt a quantitative fit of our real-space snapshots to the exact sech^2 soliton profile, Eq. (20); this would be a natural next step to make the KdV correspondence of Section 6 quantitative rather than qualitative.

8 Conclusion

We have numerically reproduced the Fermi–Pasta–Ulam–Tsingou paradox in a chain of $N = 32$ and $N = 64$ classical anharmonic oscillators, observing near-perfect recurrence of energy to the initial normal mode for weak-to-moderate cubic (α) and quartic (β) nonlinearity, and a systematic degradation of this recurrence into broadband, apparently ergodic energy exchange as α , β , and N increase. We derived, both in standard textbook form and specialized to the exact discretization of our simulation code, the continuum-limit reduction of the FPUT lattice to the Korteweg–de Vries equation (for α -dominated chains), the modified KdV equation (for β -dominated chains), and the Gardner equation (for the general mixed case) — all completely integrable, soliton-bearing PDEs whose near-elastic soliton dynamics explains the observed recurrence as a consequence of near-integrability rather than a fundamental breakdown of statistical mechanics. Our results are consistent with, and provide a self-contained pedagogical illustration of, the resolution of the FPUT paradox established by Zabusky and Kruskal and refined by the subsequent five decades of work on near-integrable Hamiltonian lattices.

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